

Major Project Batch No.	:	B-8		
Academic Year	:	Jul-Dec, Odd Semester, 2025-26		
Semester & Section	:	V Sem, Credits – 2, 100 Marks		
Branch	:	CSE (IoT CS & BCT)		
Course Name & Code	:	BIC586		
Title of the 5 th Sem BE Mini project work	:	Smart Automation System -ESP32 & Solid State Relay-Based Control		
Field/area of Major -project	:	IOT		
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Abstract -The Smart Automation System using ESP32 and Solid State Relay (SSR) enables remote and automated control of electrical appliances through IoT. The system provides wireless switching, real-time monitoring, energy efficiency, and safety. It supports Wi-Fi-based home/industry automation with low cost, reliability, and scalability. This smart solution aims to simplify daily operations while reducing power consumption and improving user convenience.

Introduction: Home automation improves convenience, energy management, and safety by enabling smart control of appliances. The ESP32 microcontroller with built-in Wi-Fi allows wireless device control, while the SSR provides noise-free high-voltage switching. This project integrates IoT and automation to operate appliances through a mobile app/web dashboard. The system aims to deliver effortless control of household and industrial devices with minimal user intervention.

Objective/Aim: To design and develop a Military Support and Rescue Bot that can effectively counter mobile robots with physical attacks while avoiding obstacles.

1. To design a smart automation system using ESP32 and SSR.
2. To remotely monitor and control electrical devices via Wi-Fi.
3. To increase safety, efficiency, and convenience in residential/industrial environments.

Problem Definition: Manual switching of electrical appliances is time-consuming, unsafe in certain environments, and prone to energy wastage. Existing automation systems are expensive or require complex installation. A cost-effective IoT-based solution is needed for reliable remote appliance control. Additionally, there is a need for a system that enables continuous monitoring of device status to prevent misuse and power loss. The solution must also be scalable so it can be adapted for both household and industrial applications.

Motivation Obtained: Growing demand for smart homes, industrial safety and energy conservation inspired this project. The availability of low-cost IoT modules like ESP32 encourages the development of affordable smart control systems.

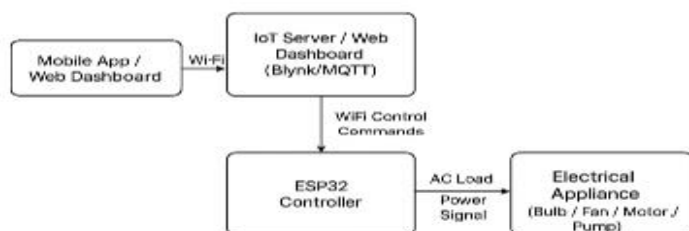
Design Concepts Used: The system is designed using IoT-based Wi-Fi control to operate appliances wirelessly, while SSR ensures safe high-voltage switching with galvanic isolation. It also provides real-time feedback of appliance status and includes an easy-to-use mobile/web interface for user control. The modular design allows the system to be easily expanded to control multiple devices simultaneously. Overall, the concept focuses on achieving automation that is low-cost, energy-efficient, and user-friendly.

Algorithm going to be used: The control algorithm follows a simple event-driven logic to ensure fast and reliable switching of appliances through IoT commands.

1. Initialize ESP32 Wi-Fi connection.
2. Connect to IoT server/dashboard.
3. Wait for the appliance ON/OFF command.
4. Trigger SSR accordingly to control the device.
5. Continuously update device status to the dashboard.

Possible Tools to be used (h/w & s/w): The hardware components involved in the system include the ESP32 development board, Solid State Relay (SSR), a power supply, a Wi-Fi router, and an electrical load such as a bulb, fan, or motor. The software requirements include the Arduino IDE along with Embedded C/Arduino libraries, and an IoT platform such as Blynk, MQTT, or a custom web server for dashboard control and communication.

Block-Diagram:



Brief working of the developed h/w or s/w module: The user sends ON/OFF commands through a smartphone application. The ESP32 receives commands via Wi-Fi and triggers the SSR, which switches the connected appliance. Feedback is sent back to the dashboard to display the current status of the device.

Expected Results (Simulation/Experimental): The system is expected to enable wireless switching of electrical appliances with a fast response time of approximately 1–2 seconds. Real-time feedback of the device status should be displayed on the IoT dashboard without delay. The Solid State Relay (SSR) is anticipated to provide safe and reliable high-voltage control while ensuring zero-noise and spark-free operation during switching.

Applications: The system can be effectively used in smart home automation to control household appliances remotely. It is also suitable for industrial appliance control, where machinery and equipment require automated switching. In smart agriculture, it can be applied to pump and irrigation system management. Additionally, it supports smart classrooms and office automation by enabling centralized control of electrical devices. Overall, the system contributes to energy-conservation applications by reducing unnecessary power consumption.

Expected Outcome of the major project: A working prototype of an IoT-based automation system that can safely control home/industrial appliances using ESP32 and SSR with a mobile/web interface.

Conclusions: The proposed system provides an efficient and low-cost method for smart automation. ESP32 offers reliable wireless connectivity, and SSR ensures safe switching, making the design scalable for real-world home and industrial applications.

References used (if any):

1. ESP32 Technical Manual – Espressif Systems
2. SSR Datasheet – Omron / Fotek
3. Blynk Documentation for IoT Device Control
4. MQTT Protocol Specification

Appendix (Datasheets / Codes) if any: (IC handbooks)

1. July 2025: Topic selection, problem identification, case study review, and confirmation of project guide.
2. August 2025: Finalization of project title, approval from guide, and collection of research/literature papers.
3. September 2025: Begin minor project work — study, design, and development of software module.
4. October 2025: Prototype development — hardware module assembling, testing, and integration with software, followed by trial runs.
5. November 2025: Final implementation, results evaluation, documentation completion, and full project review.
6. December 2025: SEE exam, project viva, paper presentation, and exhibition/demo of hardware + software model.